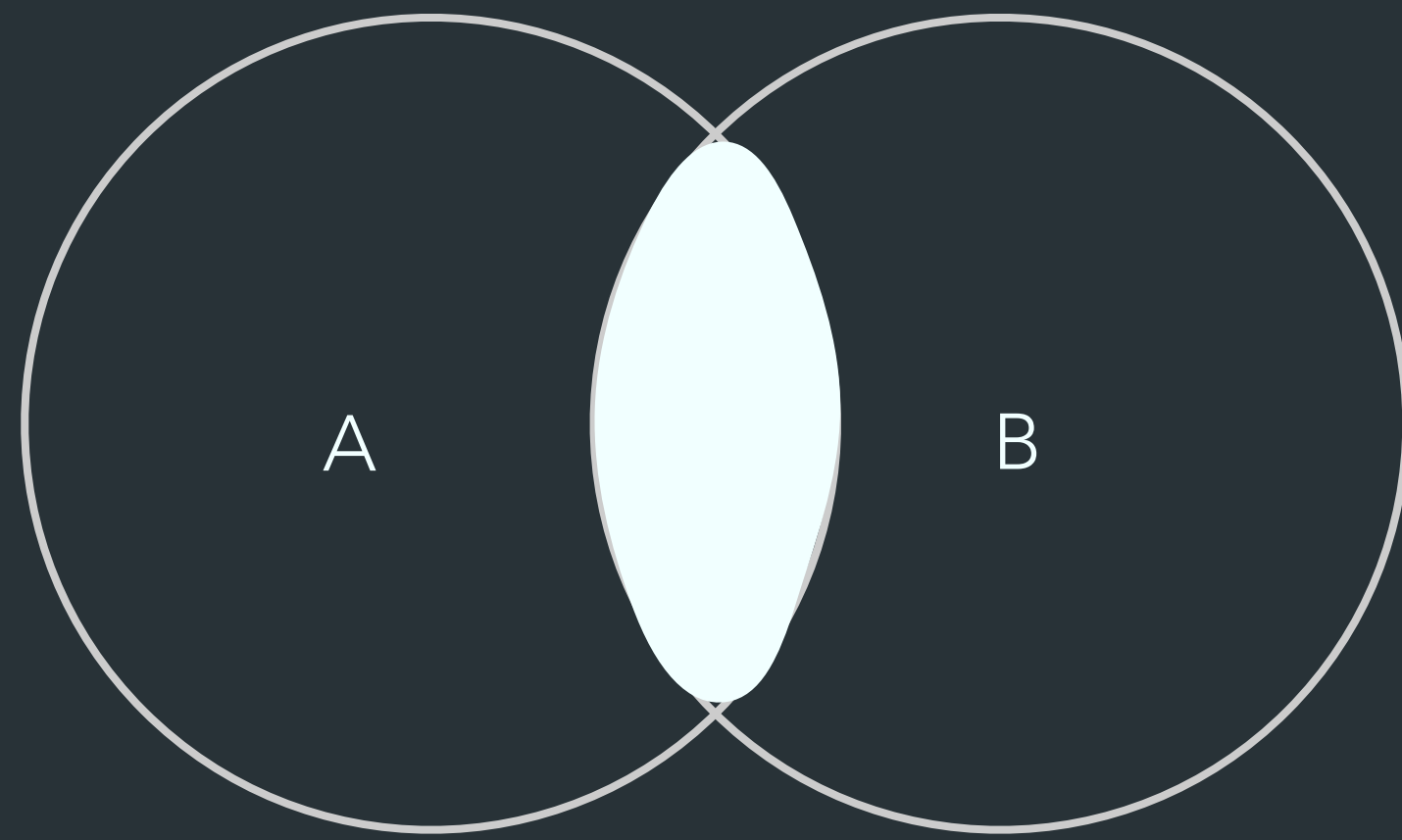




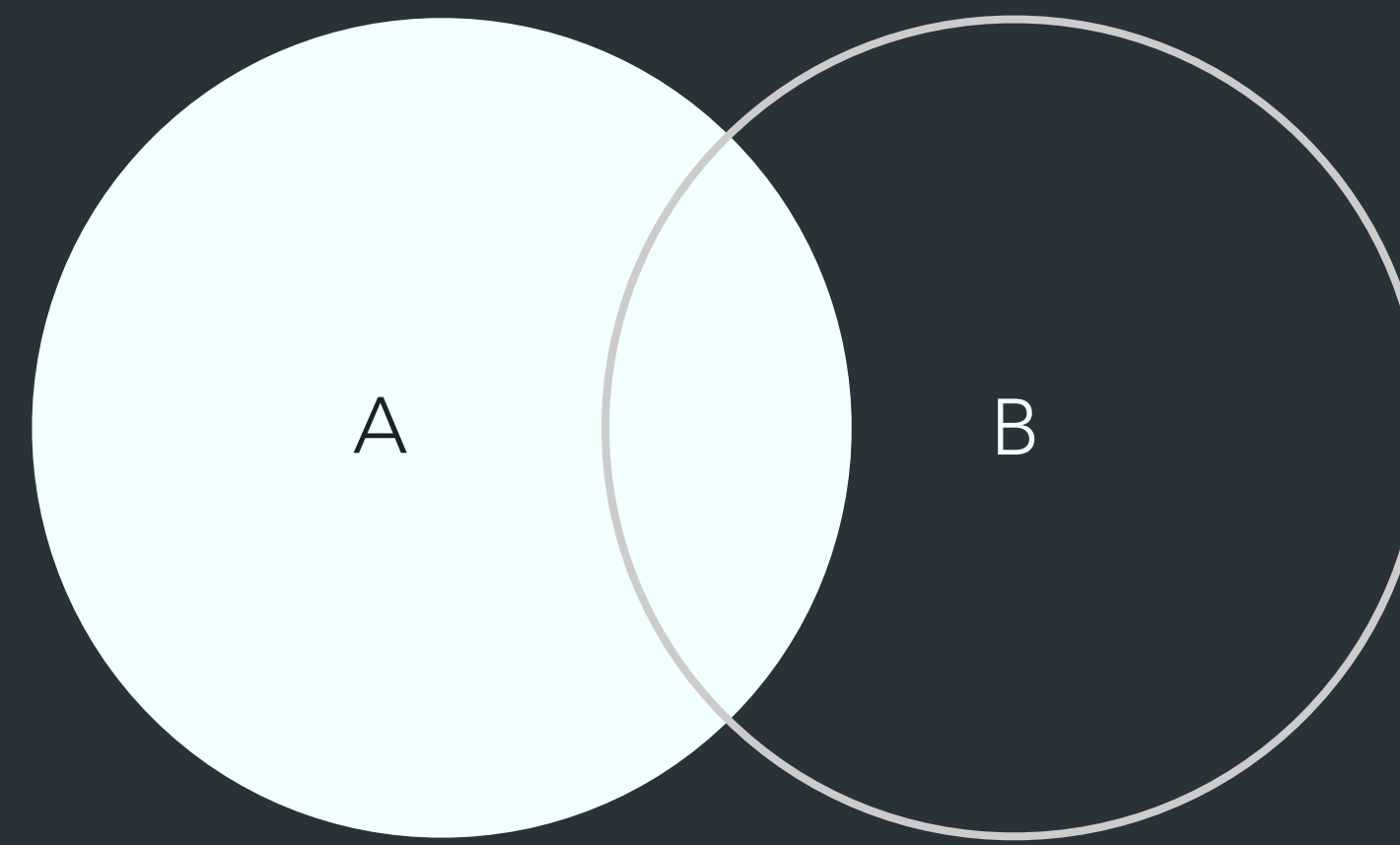
kokchun giang

working with **joins**
in sql to combine
columns from one
or more tables

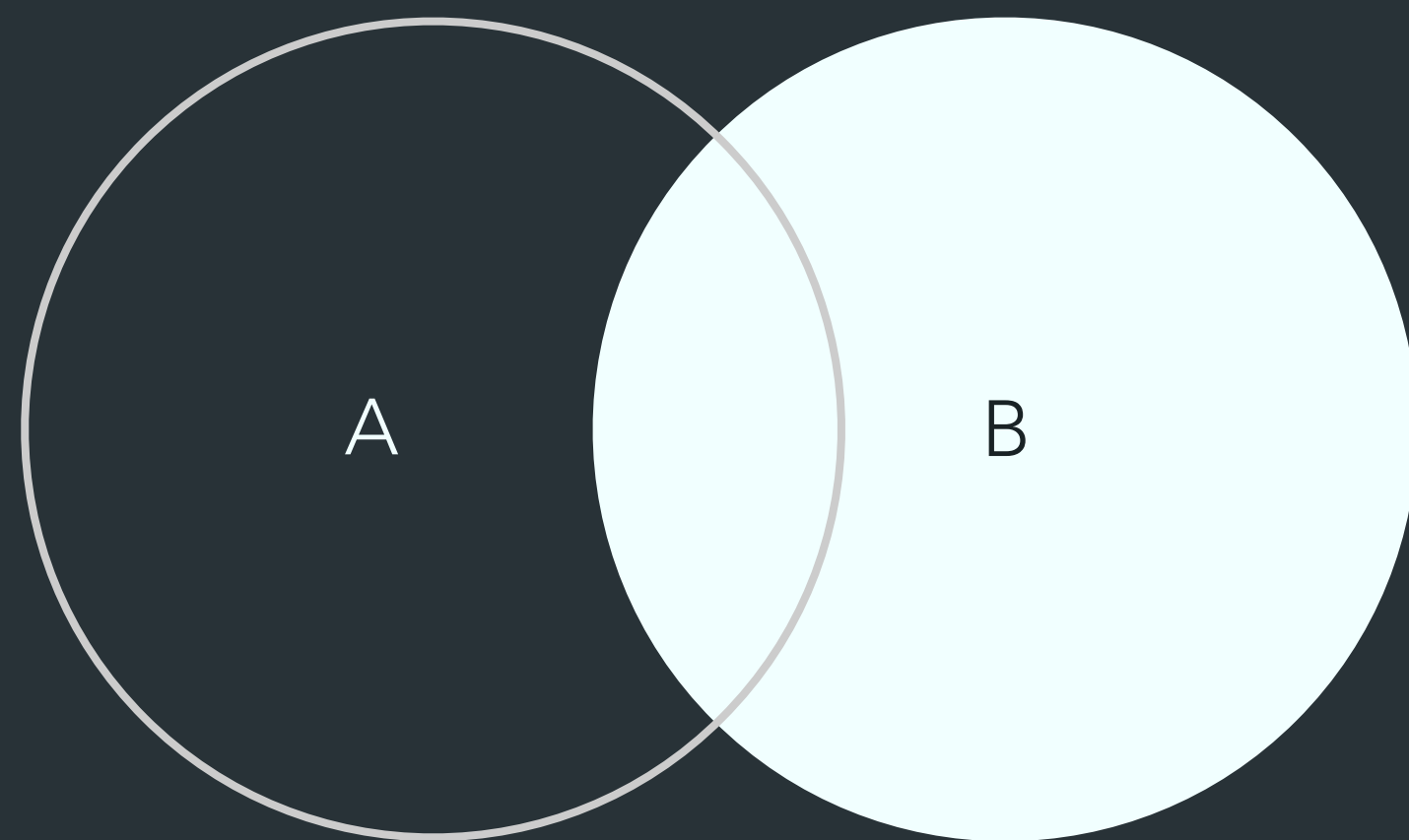
joins in SQL explained with **venn diagrams**



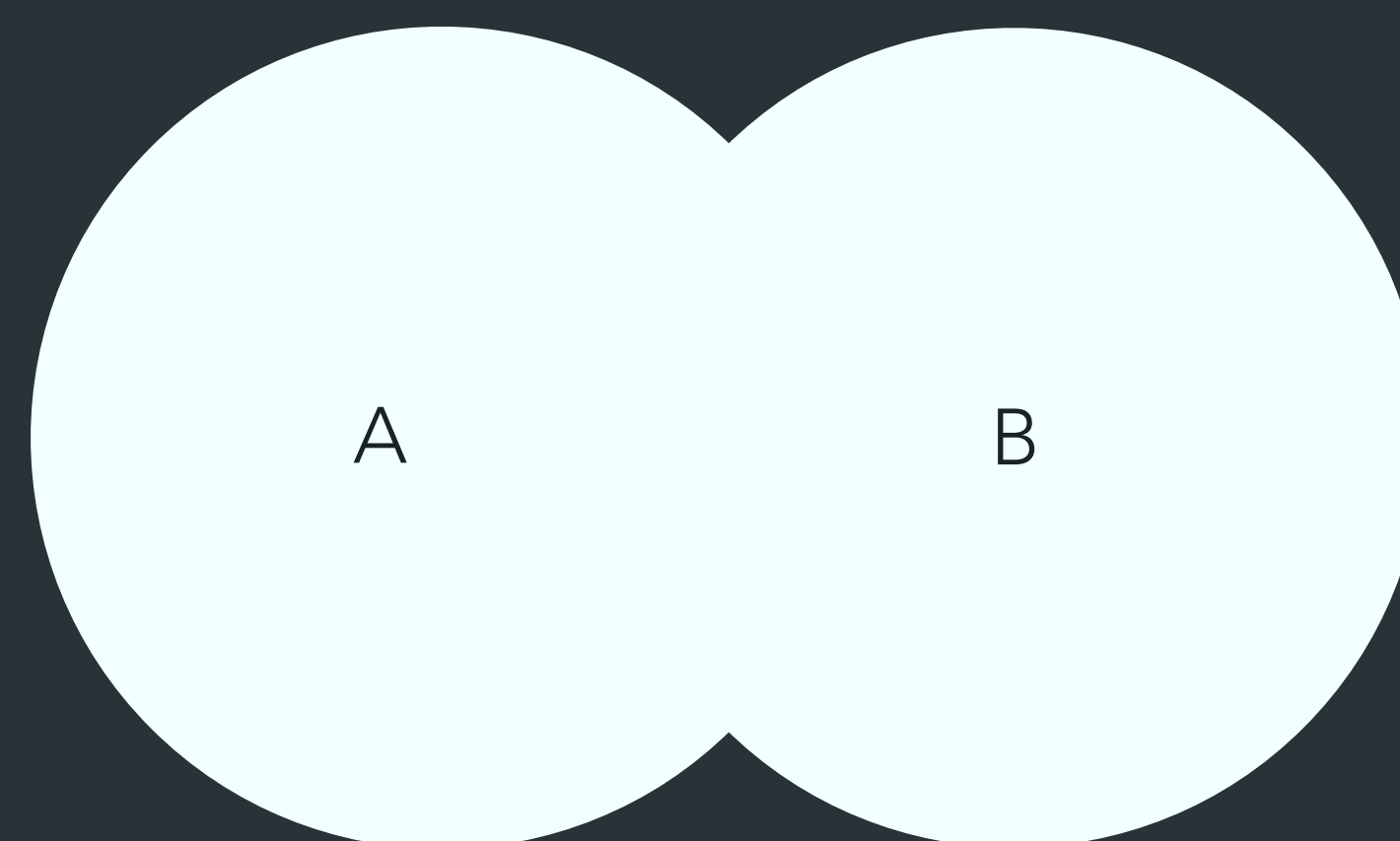
INNER JOIN



LEFT JOIN



RIGHT JOIN



FULL JOIN

here A and B are
tables and it's the
columns that are
joined

combine two tables with LEFT JOIN

```
SELECT
  p.plant_id,
  p.plant_name,
  p.type,
  pc.water_schedule,
  pc.sunlight
FROM
  main.plants p
LEFT JOIN
  main.plant_care pc ON p.plant_id = pc.plant_id;
```

matching column

returns all records from left table and match with right table, giving NULLs in right columns if there is no match

plant_id int32	plant_name varchar	type varchar
1	Rose	Flower
2	Oak	Tree
3	Tulip	Flower
4	Cactus	Succulent
5	Sunflower	Flower

plants

id int32	plant_id int32	water_schedule varchar	sunlight varchar
1	1	Daily	Full Sun
2	3	Weekly	Partial Sun
3	4	Biweekly	Full Sun
4	6	Daily	Shade

plant_care

LEFT JOIN

plant_id int32	plant_name varchar	type varchar	water_schedule varchar	sunlight varchar
1	Rose	Flower	Daily	Full Sun
3	Tulip	Flower	Weekly	Partial Sun
4	Cactus	Succulent	Biweekly	Full Sun
2	Oak	Tree		
5	Sunflower	Flower		

NULLs

combine two tables with **RIGHT JOIN**

```
SELECT
  p.plant_id,
  p.plant_name,
  p.type,
  pc.water_schedule,
  pc.sunlight
FROM
  main.plants p
RIGHT JOIN
  main.plant_care pc ON p.plant_id = pc.plant_id;
```

returns all records from right table and match with left table, giving NULLs in left columns if there is no match

plant_id int32	plant_name varchar	type varchar
1	Rose	Flower
2	Oak	Tree
3	Tulip	Flower
4	Cactus	Succulent
5	Sunflower	Flower

plants

id int32	plant_id int32	water_schedule varchar	sunlight varchar
1	1	Daily	Full Sun
2	3	Weekly	Partial Sun
3	4	Biweekly	Full Sun
4	6	Daily	Shade

plant_care

RIGHT JOIN

plant_id int32	plant_name varchar	type varchar	water_schedule varchar	sunlight varchar
1	Rose	Flower	Daily	Full Sun
3	Tulip	Flower	Weekly	Partial Sun
4	Cactus	Succulent	Biweekly	Full Sun
			Daily	Shade

NULLs

combine two tables with **INNER JOIN**

```
SELECT
  p.plant_id,
  p.plant_name,
  p.type,
  pc.water_schedule,
  pc.sunlight
FROM
  main.plants p
INNER JOIN
  main.plant_care pc ON p.plant_id = pc.plant_id;
```

matching column



returns all records where there is a match in both tables , hence no NULLs

plant_id int32	plant_name varchar	type varchar
1	Rose	Flower
2	Oak	Tree
3	Tulip	Flower
4	Cactus	Succulent
5	Sunflower	Flower

plants

id int32	plant_id int32	water_schedule varchar	sunlight varchar
1	1	Daily	Full Sun
2	3	Weekly	Partial Sun
3	4	Biweekly	Full Sun
4	6	Daily	Shade

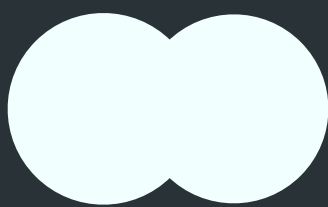
plant_care

INNER JOIN



plant_id int32	plant_name varchar	type varchar	water_schedule varchar	sunlight varchar
1	Rose	Flower	Daily	Full Sun
3	Tulip	Flower	Weekly	Partial Sun
4	Cactus	Succulent	Biweekly	Full Sun

combine two tables with **FULL JOIN**



```
SELECT
  p.plant_id,
  p.plant_name,
  p.type,
  pc.water_schedule,
  pc.sunlight
FROM
  main.plants p
FULL JOIN
  main.plant_care pc ON p.plant_id = pc.plant_id;
```

returns all records from both tables,
if there's no match, NULLs are used
on non-matching table

plant_id int32	plant_name varchar	type varchar
1	Rose	Flower
2	Oak	Tree
3	Tulip	Flower
4	Cactus	Succulent
5	Sunflower	Flower

plants

id int32	plant_id int32	water_schedule varchar	sunlight varchar
1	1	Daily	Full Sun
2	3	Weekly	Partial Sun
3	4	Biweekly	Full Sun
4	6	Daily	Shade

plant_care

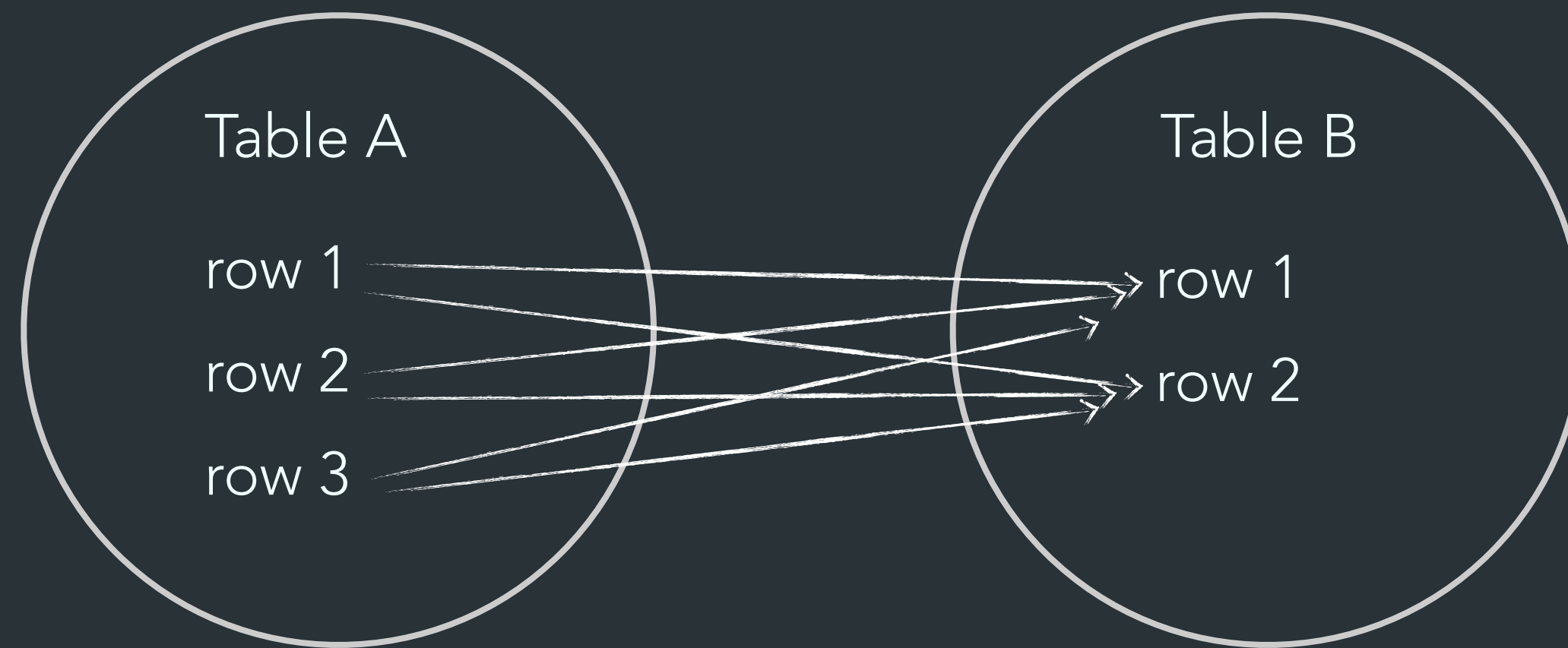
FULL JOIN

plant_id int32	plant_name varchar	type varchar	water_schedule varchar	sunlight varchar
1	Rose	Flower	Daily	Full Sun
3	Tulip	Flower	Weekly	Partial Sun
4	Cactus	Succulent	Biweekly	Full Sun
2	Oak	Tree		
5	Sunflower	Flower		
			Daily	Shade

NULLs

NULLs

cross join in SQL explained with **venn diagrams**



cross join in SQL explained with **venn diagrams**

```
SELECT
  p.plant_id,
  p.plant_name,
  pc.water_schedule,
  pc.sunlight
FROM
  plants p
  CROSS JOIN plant_care pc;
```

plant_id int32	plant_name varchar	type varchar
1	Rose	Flower
2	Oak	Tree
3	Tulip	Flower
4	Cactus	Succulent
5	Sunflower	Flower

plants

id int32	plant_id int32	water_schedule varchar	sunlight varchar
1	1	Daily	Full Sun
2	3	Weekly	Partial Sun
3	4	Biweekly	Full Sun
4	6	Daily	Shade

plant_care



plant_id int32	plant_name varchar	water_schedule varchar	sunlight varchar
1	Rose	Daily	Full Sun
2	Oak	Daily	Full Sun
3	Tulip	Daily	Full Sun
4	Cactus	Daily	Full Sun
5	Sunflower	Daily	Full Sun
1	Rose	Weekly	Partial Sun
2	Oak	Weekly	Partial Sun
3	Tulip	Weekly	Partial Sun
4	Cactus	Weekly	Partial Sun
5	Sunflower	Weekly	Partial Sun
1	Rose	Biweekly	Full Sun
2	Oak	Biweekly	Full Sun
3	Tulip	Biweekly	Full Sun
4	Cactus	Biweekly	Full Sun
5	Sunflower	Biweekly	Full Sun

returns all records joins all rows from left table with all rows in right table